

Orlando Reyes

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Summary

Electronics engineering student focused on embedded systems and low-level software development. Hands-on experience with bare-metal STM32/ARM firmware, ESP32, FPGA/VHDL digital design, and Docker-based infrastructure. Experienced across the full stack, from register-level peripheral drivers to Linux and Python automation tooling.

Education

Engineer in Electronics — Universidad Autónoma de Aguascalientes 08/2023 – Present (Expected 2027)
Vice Chair and founding member of the IEEE Student Branch at UAA. Experienced in hardware documentation and technical publishing using LaTeX.

Technician in Mechatronics — CBTis No. 168 08/2019 – 07/2022
Hands-on training in diagnostics, maintenance of mechatronic equipment, AutoCAD drafting, and CNC programming.

Embedded & Software Projects

SCADA System for Industrial PLC — Designed and implemented a low-cost SCADA system integrating a Linux-based industrial PLC with Node-RED for real-time HMI monitoring and secure remote access via a WireGuard-based mesh VPN.

RISC-V Processor Core — Designed and implemented a custom 32-bit RISC-V processor core in VHDL for FPGA logic synthesis and functional hardware execution.

IEEE-754 Floating-Point Calculator — Developed a 32-bit floating-point arithmetic unit in VHDL and ARM Assembly featuring UART-based dynamic I/O communication.

STM32 Bare-Metal Drivers — Wrote register-level SPI drivers for ILI9341 and ST7735R TFT displays with XPT2046 resistive touch controller integration directly from datasheets.

Custom PCB & Power Hardware Design — Designed minimum-system MCU development boards in KiCad and developed power electronics topologies with custom board layouts.

FPV Flight Controller Board Debugging — Assembled an FPV quadcopter running Betaflight on an STM32F103 flight controller; resolved hardware issues on non-standard clone boards.

Self-Hosted Homelab Infrastructure — Architected and managed a self-hosted Linux NAS containerizing microservices like Home Assistant, local storage, and media streaming via Docker.

Workflow Automation & DevTools — Built Python and Playwright web scrapers/automation tools utilizing Chrome DevTools Protocol (CDP) for event management systems.

Professional Experience

Database & Systems Administrator — Academia Yamaha AGS 05/2024 – 06/2026

- Standardized student databases and automated reporting workflows using custom Python scripts and VBA for Excel.
- Managed SAT-compliant electronic invoicing and inventory tracking software tools.

Team Leader IS-Warehouse — Sensata Technologies 01/2023 – 03/2024

- Built automated reporting models in Excel for real-time production tracking and schedule monitoring.
- Coordinated material logistics and stock availability based on high-volume production demand.

Technical Skills

Programming: C, C++, VHDL, ARM Assembly, Python, VBA

Embedded Systems: STM32 (Cortex-M), ESP32/ESP8266, PIC, AVR

Protocols: UART/USART, SPI, I2C, CAN

Tools & OS: KiCad, Git, Docker, Linux, STM32CubeIDE, PlatformIO

Languages

English: B1 / Intermediate (Professional Reading & Writing)

Soft Skills

Team Leadership, Technical Writing, Project Coordination, Troubleshooting